

From Manuscript to Pixel: Tips for a Digitizing Campaign and for Handling Endangered Sources

Digitization of Polyphonic Choirbooks in Guatemala

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“Musicologists and Their Sources:
The Alchemy of Turning Parchment into Pixels”

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Outline

- Guatemalan Cathedral (GuatC) polyphonic choirbook collection
- Digitization of GuatC 1
- Final remarks

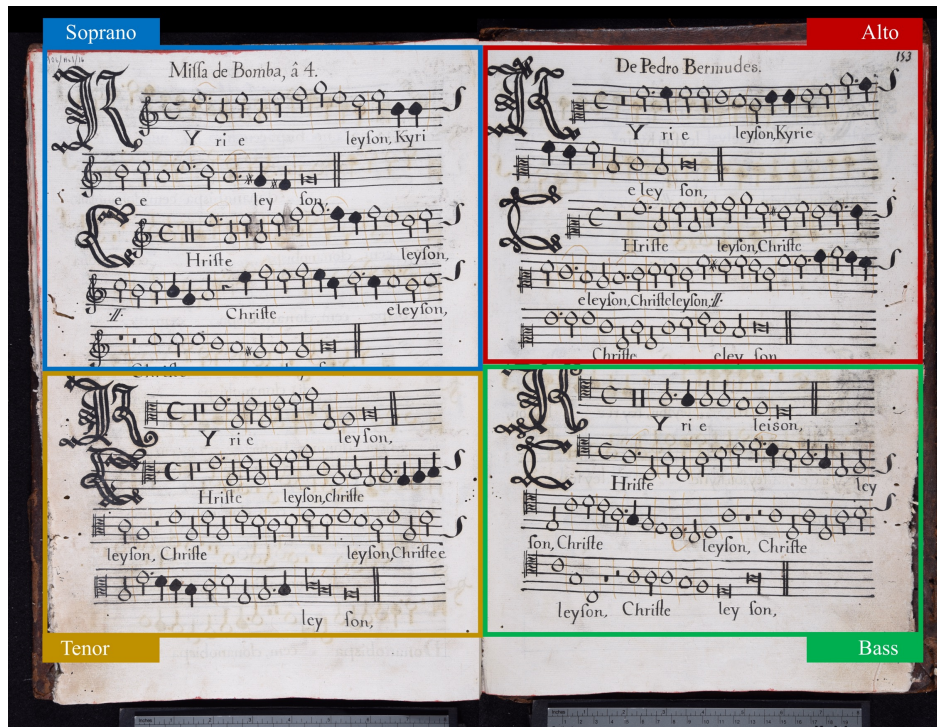
The GuatC Collection

Collection of six choirbooks, held at the *Archivo Histórico Arquidiocesano de Guatemala* (Archdiocesan Historical Archive of Guatemala, AHAG) in Guatemala City.



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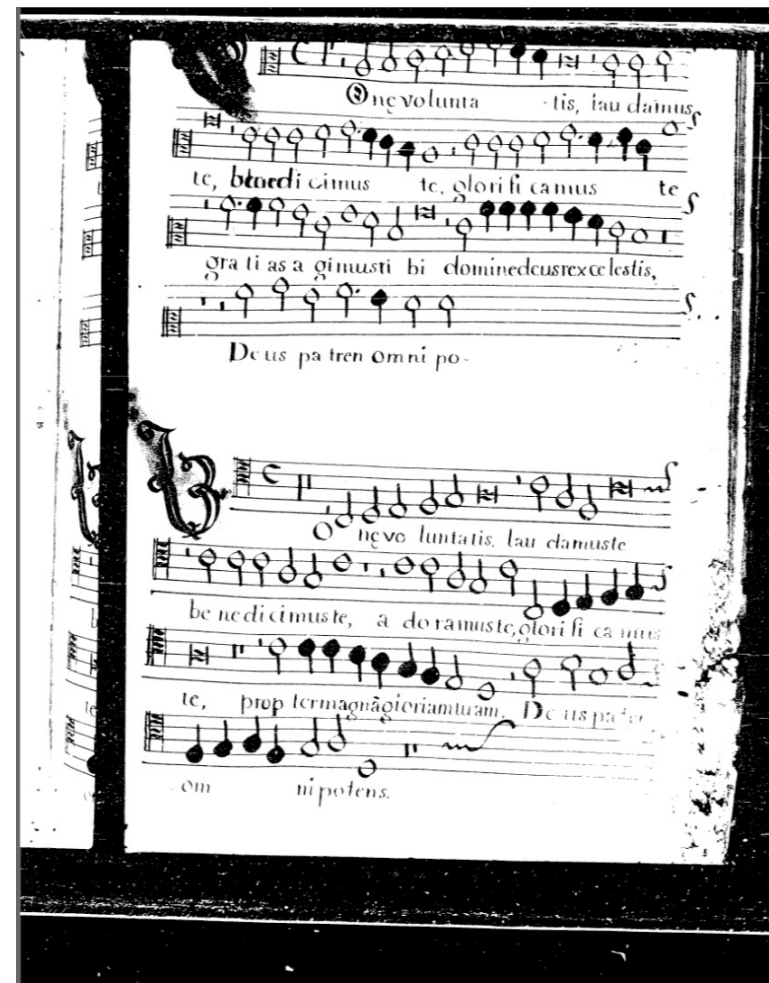
- Manuscripts written in paper
- Large dimensions
 - Average folio size = 30 cm x 45 cm
 - Book 6, size = 21.5 cm x 30 cm
- In mensural notation
- Voices in choirbook format

The GuatC Collection

- Copied in the 17th and 18th centuries
- Repertoire: mostly 16th c. polyphonic music
- Document a continuous performance tradition of sacred choral music from the Renaissance to early 19th century

The GuatC Collection

- Inventories of books 1 to 3:
 - Pujol (1965), Stevenson (1970), Snow (1996)
- General overview of the repertoire
 - Morales Abril (2015)
- Transcriptions into modern notation
 - Snow's transcription of 4th book
 - Dieter Lehnhoff and Omar Morales Abril
- Microfilms (1984) → Books 1 to 3
- Access → Permission from the archive



Microfilm at Centro de Investigaciones Regionales de Mesoamérica (CIRMA).

DjVu file at Música Colonial Archive. Reel 9. Folio 2r.
(https://www.cpd.org/wiki/index.php/M%C3%BAsica_Colonial_Archive)

Preserve and increase access to this
music corpus by using digitization
and music-encoding technologies

Digitization of GuatC 1

The Start

- Request of permission for the project to the AHAG:
 - Contacted Guatemalan musicologist Omar Morales Abril
 - Morales Abril advised me to contact the AHAG archivist Alejandro Conde
 - Conde suggested to ask permission for a pilot project involving **only one** of the choirbooks
 - Wrote a letter to the chancellor of the Ecclesiastical Curia of Santiago de Guatemala and director of the AHAG, Father Eddy René Calvillo, to ask permission for the pilot (23 October 2018)
 - The archivist hand it to Father Calvillo
 - Approved on 31 October 2018

The Survey & Findings

- The AHAG does not have its own digitization and conservation departments
- Outsource digitization and conservation
- **Paper conservation:**
 - Centro de Rescate, Estudio y Análisis Científico del Arte (CREA)
 - Non-profit institution
 - Conservation assessment of the six choirbooks (November 2018)

The Survey & Findings

- **Digitization:**

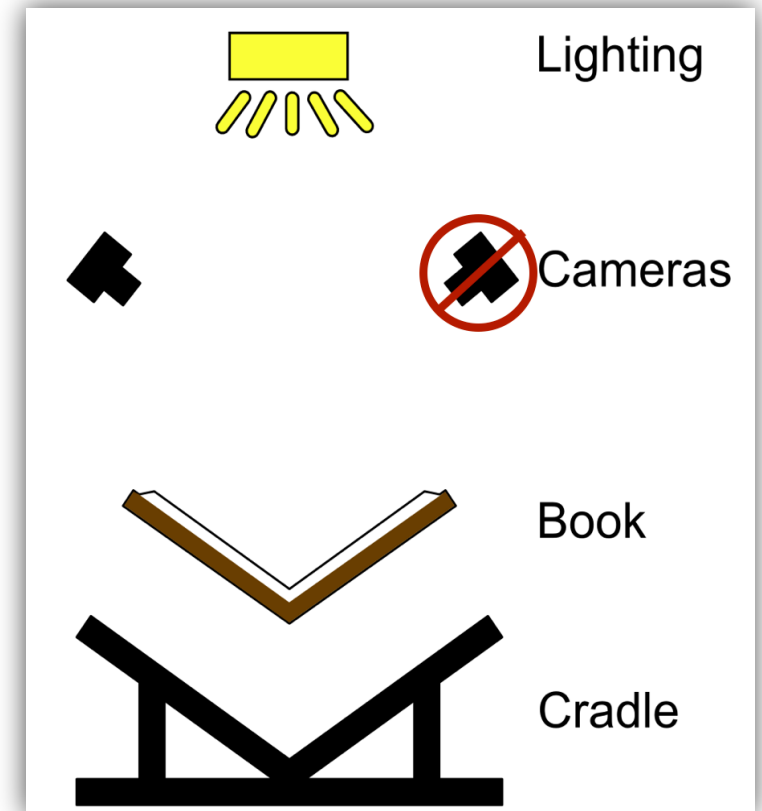
- Survey for digitization equipment and services in Guatemala City
- Nothing suitable due to the large dimensions or conservation status of the manuscripts
- Do It Yourself (DIY) Book Scanner
- No digitization technician with experience with special collections
→ Professional photographer (suggested by another archive)

Digitization of GuatC 1

THE EQUIPMENT

DIY Book Scanner

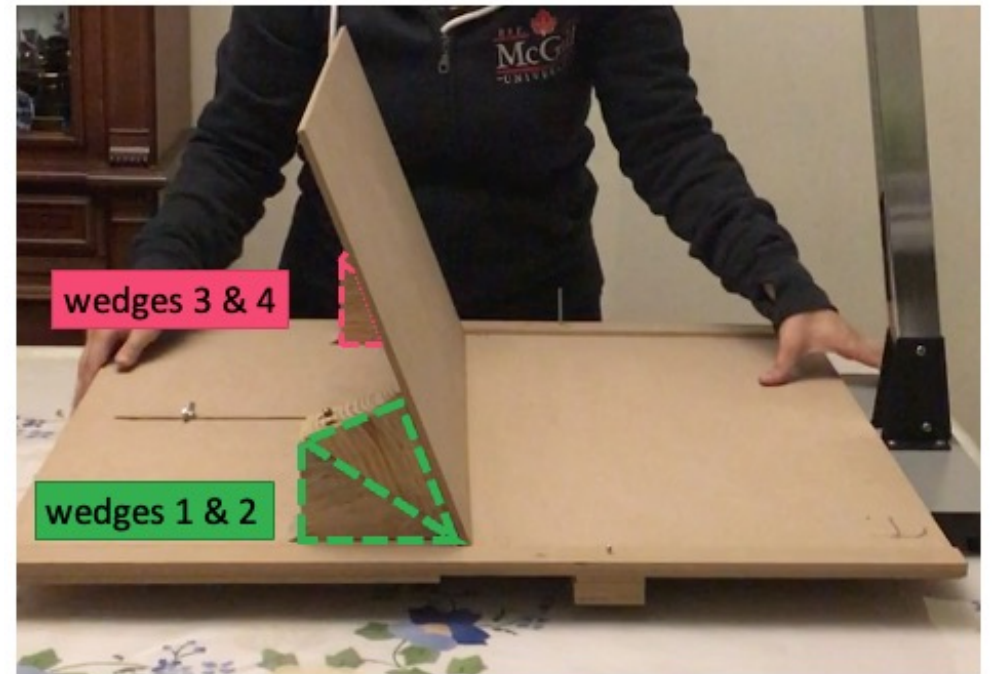
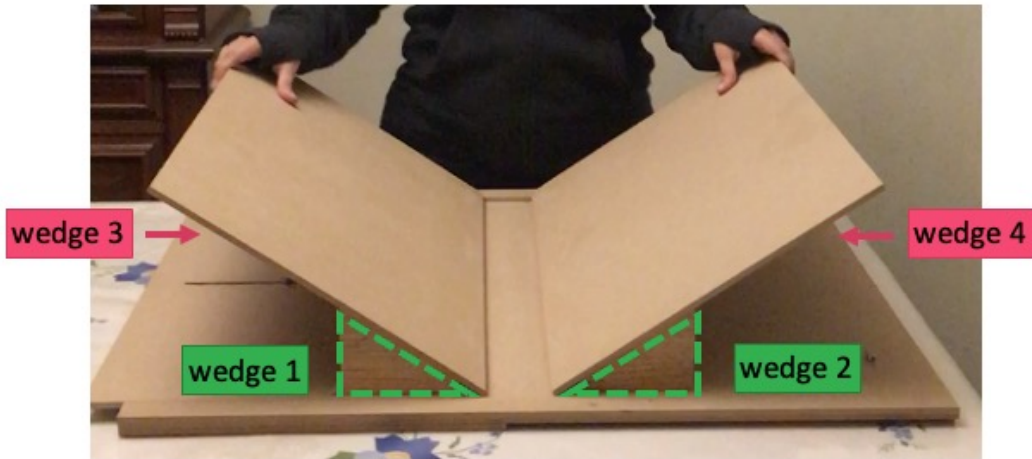
- Book cradle
 - **Built** by German Thomae (my father)
- Camera(s)
 - **Borrowed** from the McGill Music Library
 - Nikon D750 (Image sensor of 24.3-effective-megapixels)
 - Images saved at **300-dpi** resolution (with no interpolation)
- Lighting
 - **Borrowed** from DDMAL Lab, McGill University
 - Two natural-light LED panels



<http://diybookscanner.org>

Book Cradle: Two Configurations

Both allow for the book to be opened at 110° angle
By using four wedges of 35°





The Book Cradle (with copy stand)

-
- Easier to set the camera **perpendicular** to the page
 - And to keep the camera at a **constant distance** from the page

Cradle & Camera



Lighting



- Two natural-light LED panels
- Not recommended to use flash or natural light
- One at each side → even distribution
- Angle → avoid reflections on the camera lens



DIY Book Scanner

Advice from:

- Digital Image Archive of Medieval Music (DIAMM)
 - Julia Craig-McFeely
- Bibliothèque et Archives Nationales du Québec (BAnQ)
- McGill Library

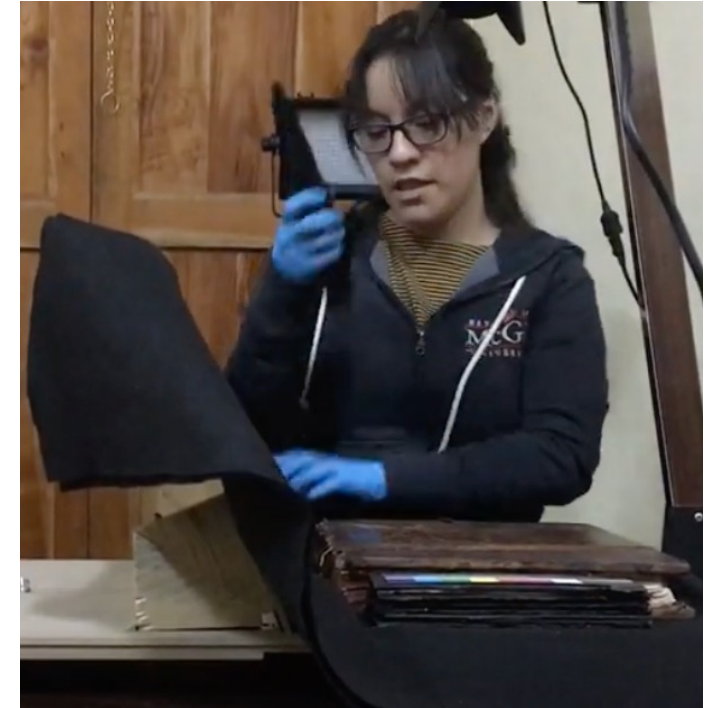
Canadian institutions

Digitization of GuatC 1

THE PROCESS

Digitization Process & Issues

- The photographer set the camera parameters on the first day
 - Since I conducted the digitization on my own, he set the camera to shoot in one of its auto modes (Mode A – Aperture-Priority Auto)
 - Also, auto white balancing & auto-focus
- Have to come up with a [small workflow \(series of steps\)](#) to make the digitization process as consistent as possible
- The “page falling” and the “increase in curvature” issues when [reaching the other end of the book](#)



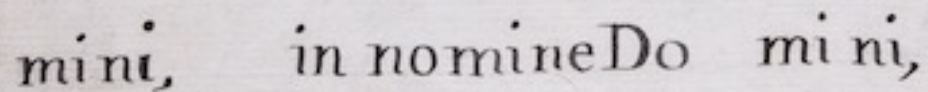
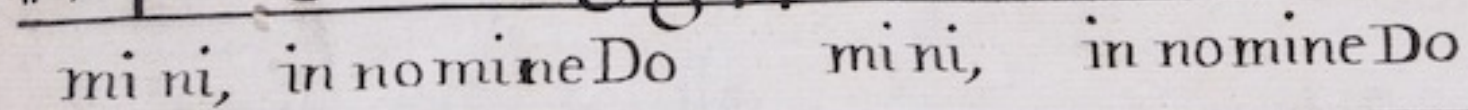
Digitization Process & Issues

- Shooting tethered with the camera connected wirelessly to my computer:
 - Preview the photos on a larger monitor
 - Shoot remotely
- Issues with computer application tethered to the camera
 - **Solution during digitization:** Switched to mobile tethered application
 - **In the future:** camera's official tethering app, connect through a (long) cable, and save images on external hard drive

Digitization Process & Issues

- As with most manuscripts, the verso of the pages is not numbered (only the front of the pages are numbered)
- Difficult to keep track of the page number for versos
- Although the tethering app can do that
- **Solution** proposed by the photographer:
Include the verso numbers in the photograph itself



iii. Derivations of $\mathcal{L}$ from $\mathcal{L}_0$ 

A Kodak Color Control Patches chart used for color calibration in photography. The top features a ruler with inches (0-8) and centimeters (0-19). Below the ruler, the text "KODAK Color Control Patches" is prominently displayed, followed by "© The Tiffen Company, 2000". To the right, the "Kodak" logo is shown above "LICENSED PRODUCT". The chart consists of two rows of nine color patches each. The first row includes patches labeled Blue, Cyan, Green, Yellow, Red, Magenta, White, 3/Color, and Black. The second row contains corresponding darker or lighter shades of these colors.

The Digitization Project (January 2019)

Monday	Tuesday	Wednesday	Thursday	Friday
JAN 07 - Conservation treatment - Test two book cradle configurations with GuatC 1	08 - Set up book scanner & camera parameters - Test workflow to find future issues	09 - Photographing the manuscript	10 - Photographing the manuscript	11 - Photographing the manuscript
14 - Photographing the manuscript	15 - Reshooting folios that had issues	16	17	18

Master: NEFF (Nikon's RAW format) & TIFF, unmodified

Access: JPEG

Resolution: 300 dpi

Bit depth: 24 bits

From a total of approx. 580 photos taken, approx. 187 were discarded

Final Remarks

- Try to present a realistic solution for the digitization of old music books in an archive that lacks the resources and personnel for this kind of project
 - There are many music documents locked in archives around the world
 - Not all these archives have digitization and conservation departments
 - Present a case study of a digitization project conducted under such conditions
- Hope it is useful for similar initiatives

Final Remarks

- Hope it is useful for similar initiatives
- **Website:** https://martha-thomae.github.io/projects/digit_guatemala
 - **YouTube Playlist:** <https://www.youtube.com/playlist?list=PLC08PttZdHNBpl8z1MV2U3UZFNBL3jQ5g>
 - **Paper:** Martha E. Thomae, Julie E. Cumming, and Ichiro Fujinaga. 2022. Digitization of Choirbooks in Guatemala. In Proceedings of the 9th International Conference on Digital Libraries for Musicology (DLfM '22). Association for Computing Machinery, New York, NY, USA, 19–26. <https://doi.org/10.1145/3543882.3543885>

Acknowledgements

- **Omar Morales Abril**, Guatemalan musicologist that provided me with advice and information about the choirbooks
- **Alejandro Conde**, AHAG's archivist that supported and facilitated the project
- **José German Thomae Villela** for constructing the cradle
- **Darryl Cameron** for lending the copystand
- **Daniel Hernández-Salazar** for setting the camera parameters
- The following experts for their advice regarding digitization and conservation of special collections: **Julia Craig-McFeely** and **Lynda Sayce** (DIAMM), **Jessica Régimbald**, **Marie-Chantal Anctil**, and **Michel Legendre** (BAnQ), and **Gregory Houston** (McGill Library)

Acknowledgements

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- **Distributed Digital Music Archives and Libraries (DDMAL) Lab at McGill University** for lending the lights and other parts of the equipment
- **Marvin Duchow Music Library at McGill University** for lending the camera
- **Centro de Rescate, Estudio y Análisis Científico del Arte (CREA)** for the conservation assessment of the collection and the conservation treatment of the choirbook



Thank you!

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https://martha-thomae.github.io/projects/digit_guatemala



MEASURING POLYPHONY
DIGITAL ENCODINGS OF LATE MEDIEVAL MUSIC



NATIONAL ENDOWMENT FOR THE
Humanities

SIMSSA | Single Interface for Music
| Score Searching and Analysis



Brandeis University



Social Sciences and Humanities
Research Council of Canada

Conseil de recherches en
sciences humaines du Canada

Canada



McGill



Schulich School of Music
École de musique Schulich

DDMAL

DISTRIBUTED DIGITAL MUSIC
ARCHIVES & LIBRARIES LAB



Centre for Interdisciplinary Research
in Music Media and Technology

Fonds de recherche
Société et culture

Québec



CENTRO DE ESTUDOS DE
SOCIOLOGIA E ESTÉTICA
MUSICAL

C|E|S|E|M



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UNIVERSIDADE NOVA DE LISBOA

fct

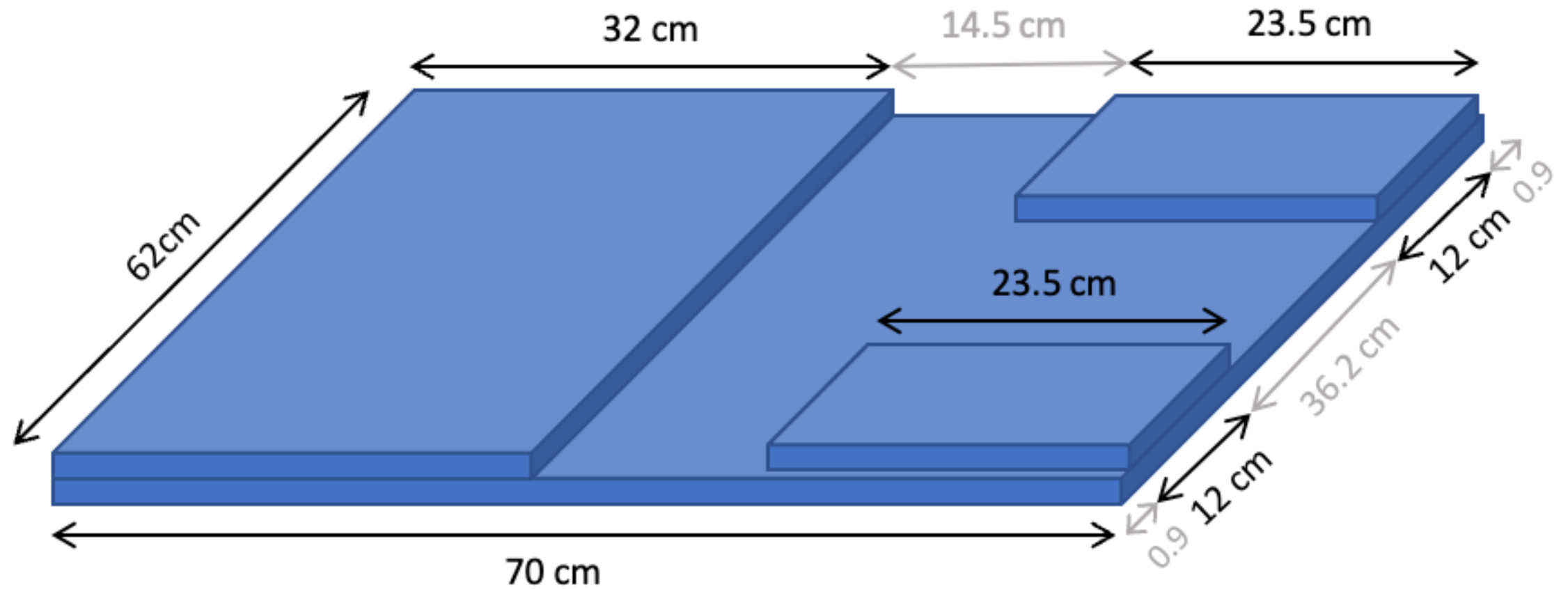
Fundação
para a Ciência
e a Tecnologia

Extra Slides

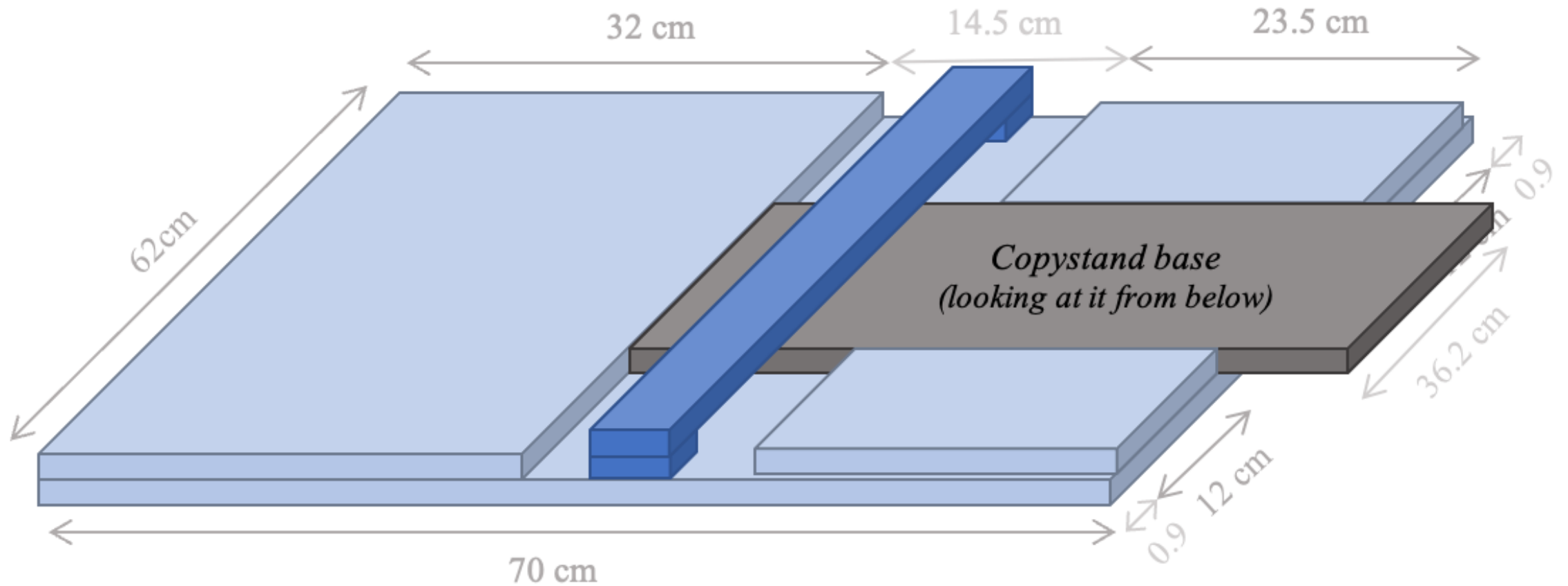
Digitization of GuatC 1

THE EQUIPMENT:
Cradle

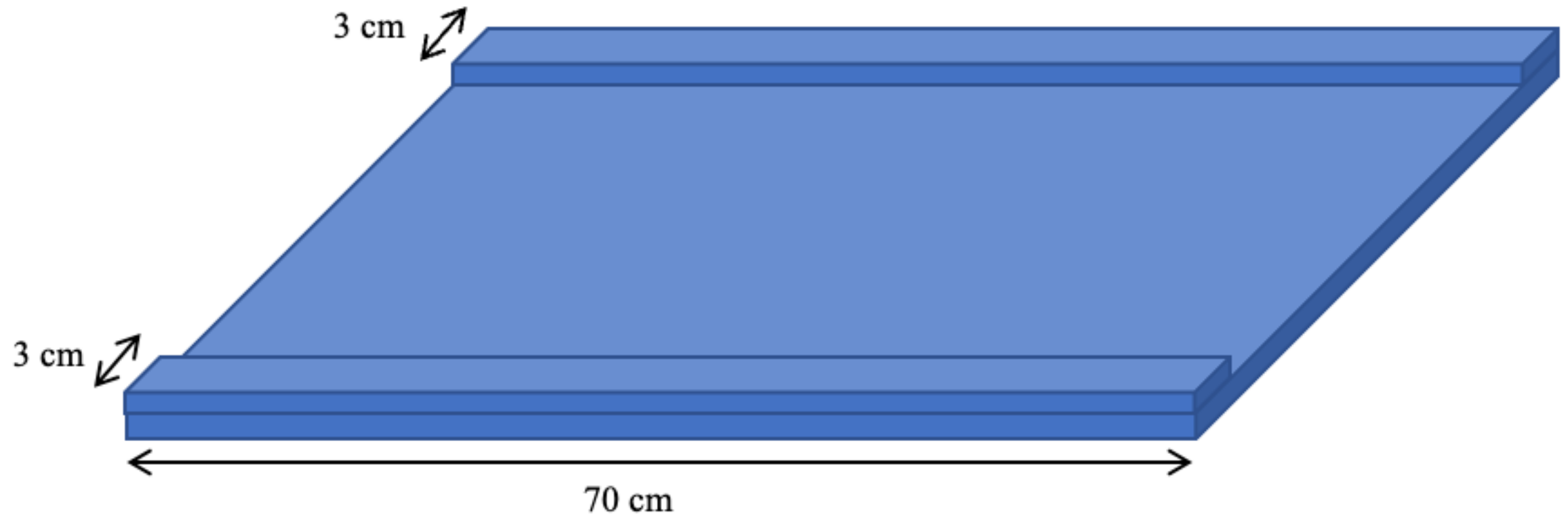
Bottom of the Cradle



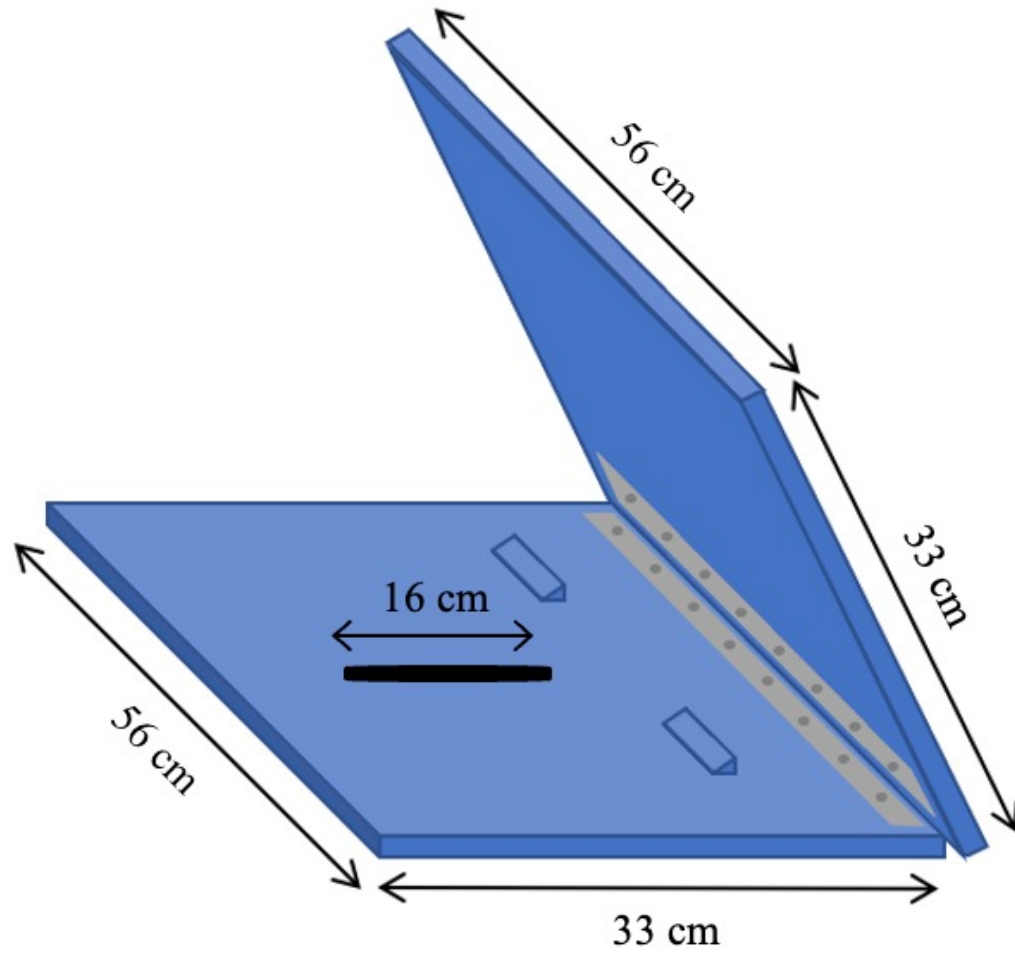
Bottom of the Cradle



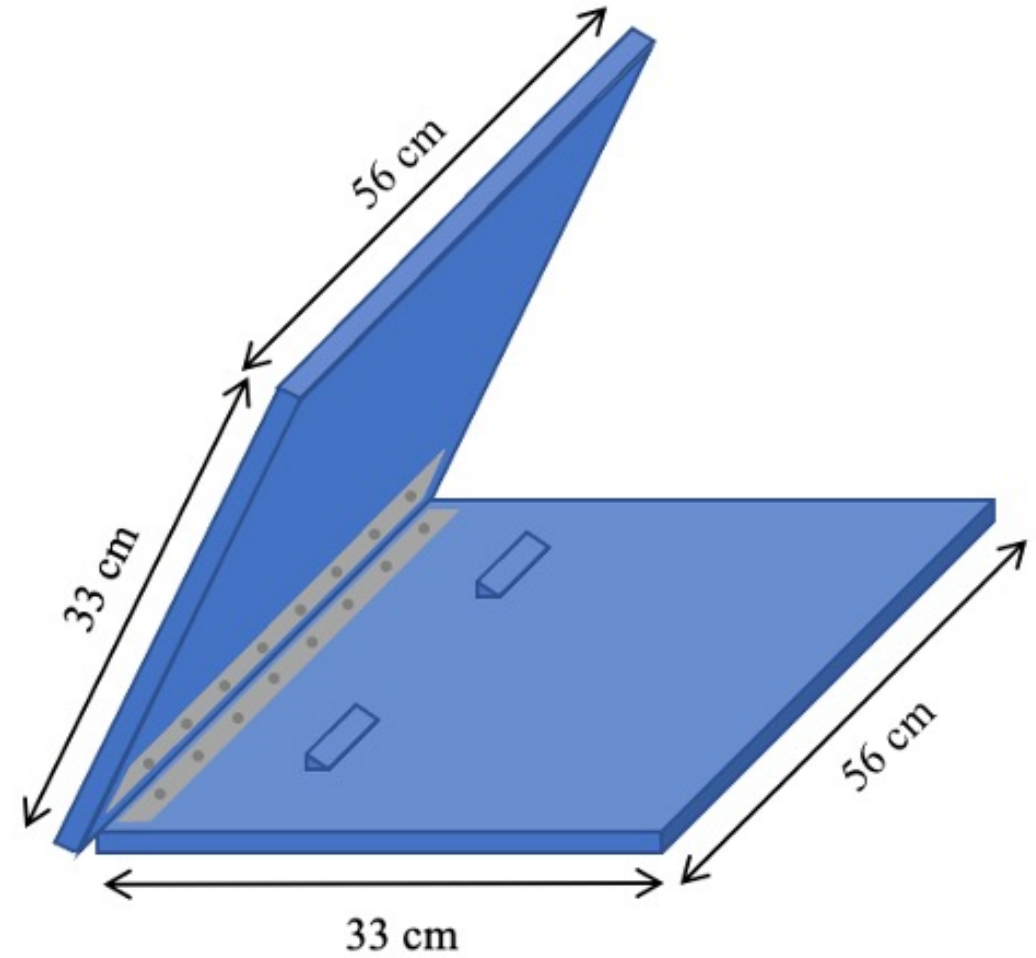
Top of the Cradle



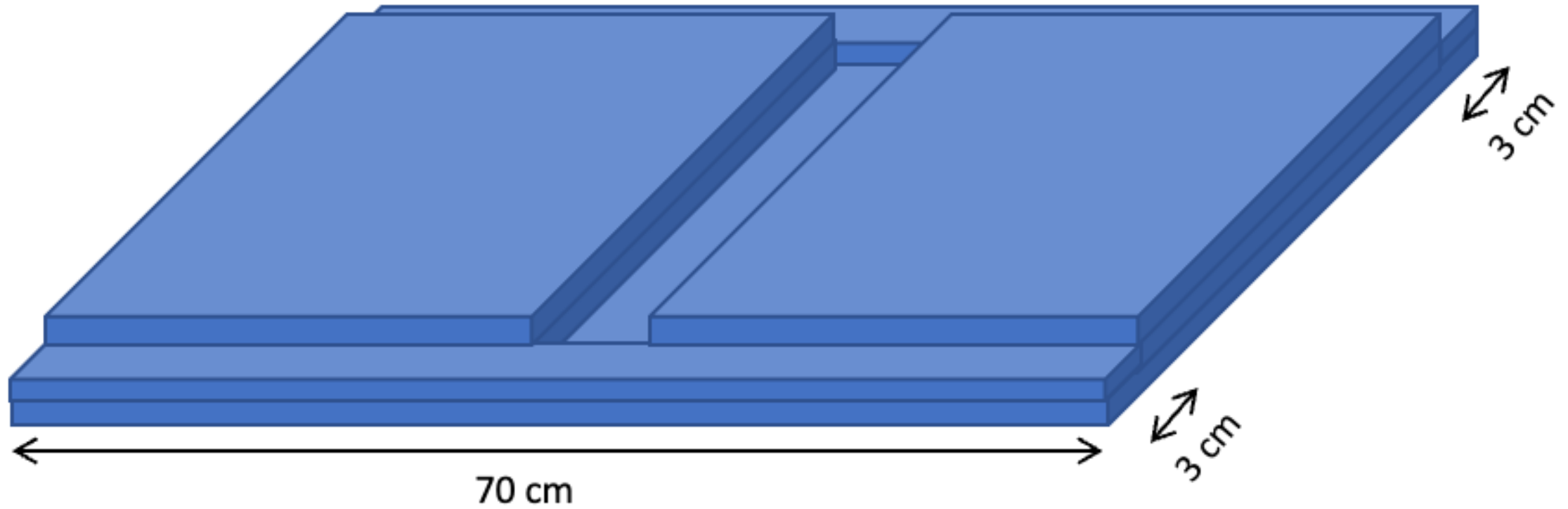
Left Door



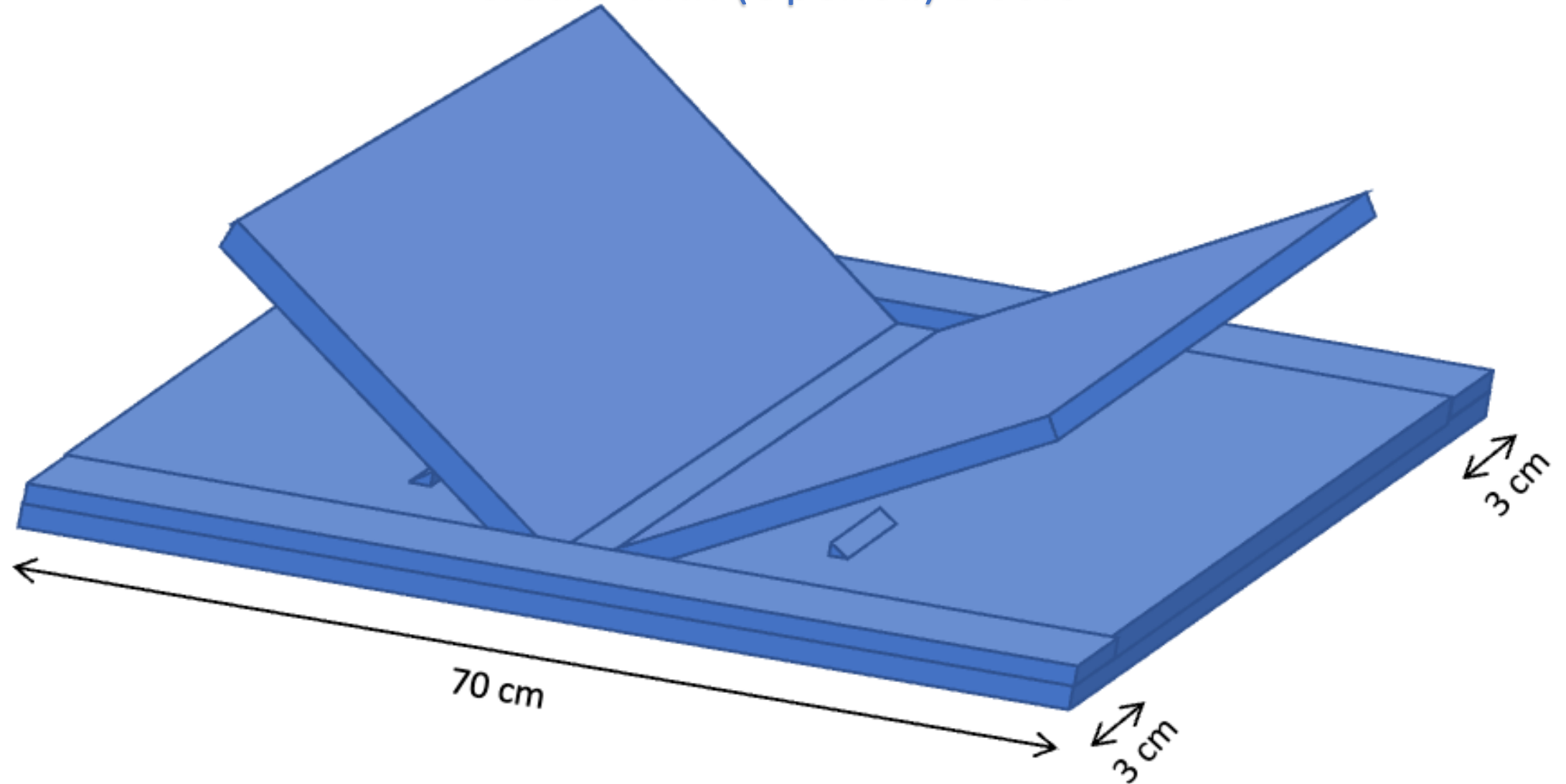
Right Door



Cradle with (Closed) Doors



Cradle with (Opened) Doors



Digitization of GuatC 1

NEXT STEPS

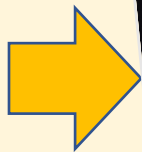
Digitization

Do-it-yourself (DIY) book scanner

Digitization



Manuscript



Digital images

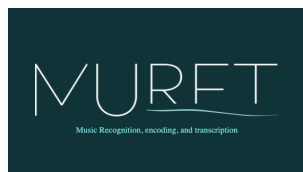
Part of a larger workflow to preserve and increase access to this music corpus through digital images and symbolic scores

Three-Step Workflow

Digitization



Optical music recognition (OMR)



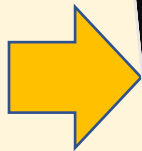
Automatic voice alignment & editorial correction

MEASURING POLYPHONY

DIGITAL ENCODINGS OF LATE MEDIEVAL MUSIC



Manuscript



Digital images

Three-Step Workflow

Digitization



Optical music recognition (OMR)



Automatic voice alignment & editorial correction

MEASURING POLYPHONY

DIGITAL ENCODINGS OF LATE MEDIEVAL MUSIC



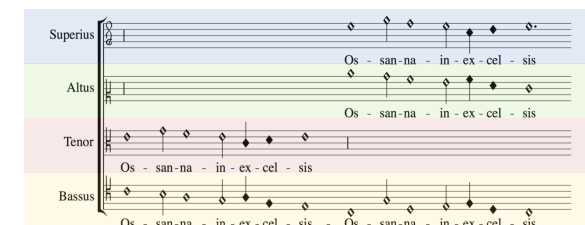
Manuscript



Digital images

```
<mei>
<music>
<staff n="1">
<clef line="2" shape="G"/>
<note dur="semibrevis" pname="g" oct="5"/>
<note dur="semibrevis" pname="d" oct="5"/>
<dot/>
<note dur="minima" pname="c" oct="5"/>
<ligature form="recta">
<note dur="semibrevis" pname="d" oct="5"/>
<note dur="semibrevis" pname="g" oct="4"/>
</ligature>
<note dur="brevis" pname="b" oct="4"/>
</staff>
</music>
</mei>
```

MEI Parts file
encodes the **symbols** in each
part/voice of the images



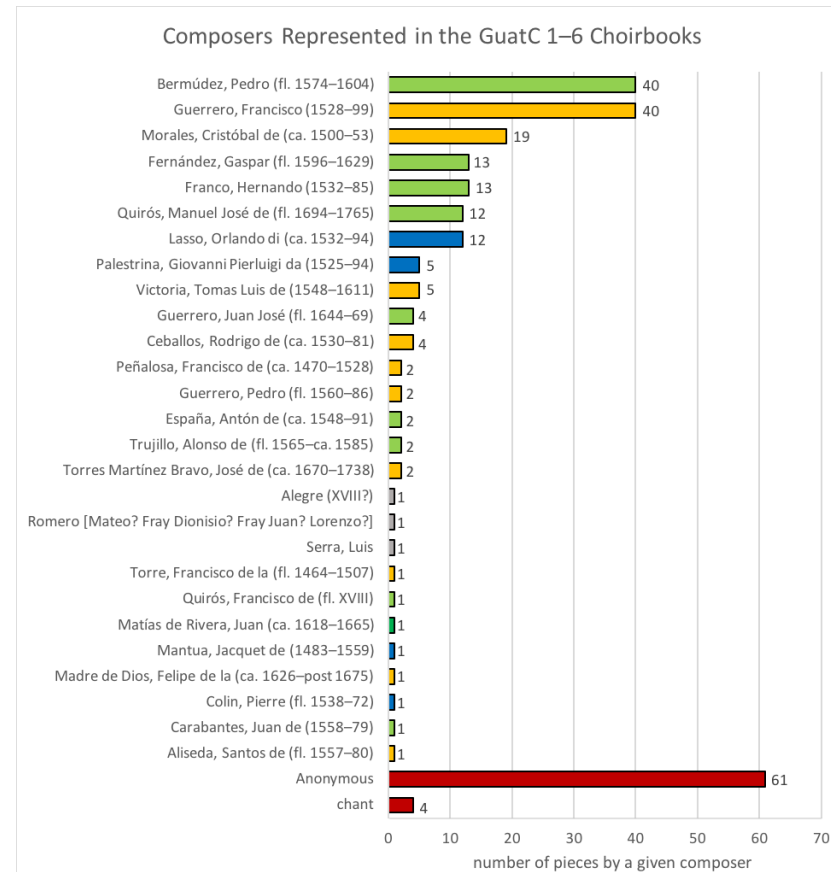
MEI Score file
interpreted durations
(i.e., *alteration & imperfection*)

The GuatC Collection

The GuatC Collection

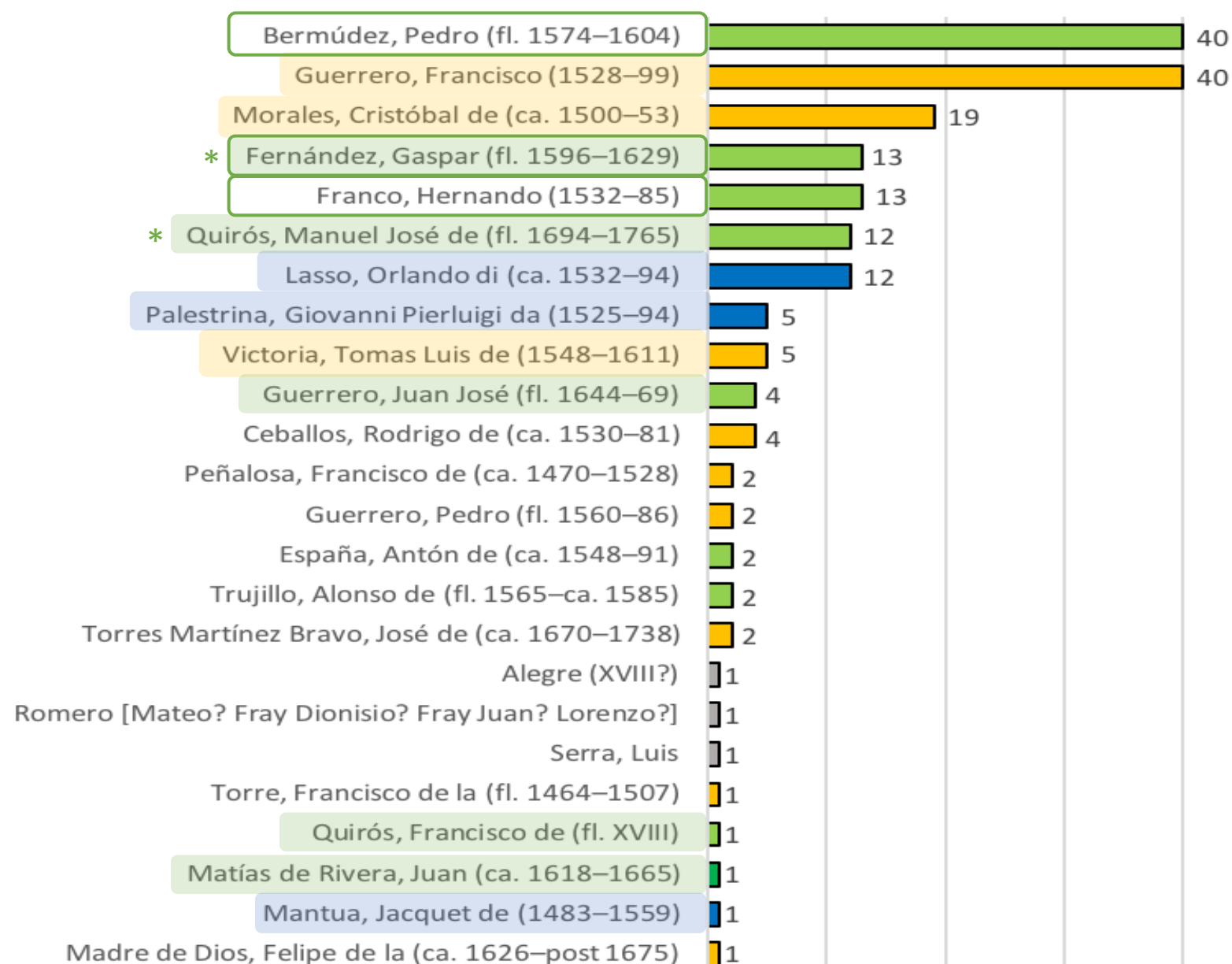
Composers	Pieces by Composer (%)
Iberian Peninsula (never set foot in the Americas and traveled to the Americas)	53%
Local composers (born and trained in Guatemala or Mexico)	12%
Other parts of Europe	8%
Region of activity is still unknown	1%
Anonymous & chant pieces	26%

The GuatC Collection



This is a graphic representation of the information provided by Morales Abril (2015, 119–120)

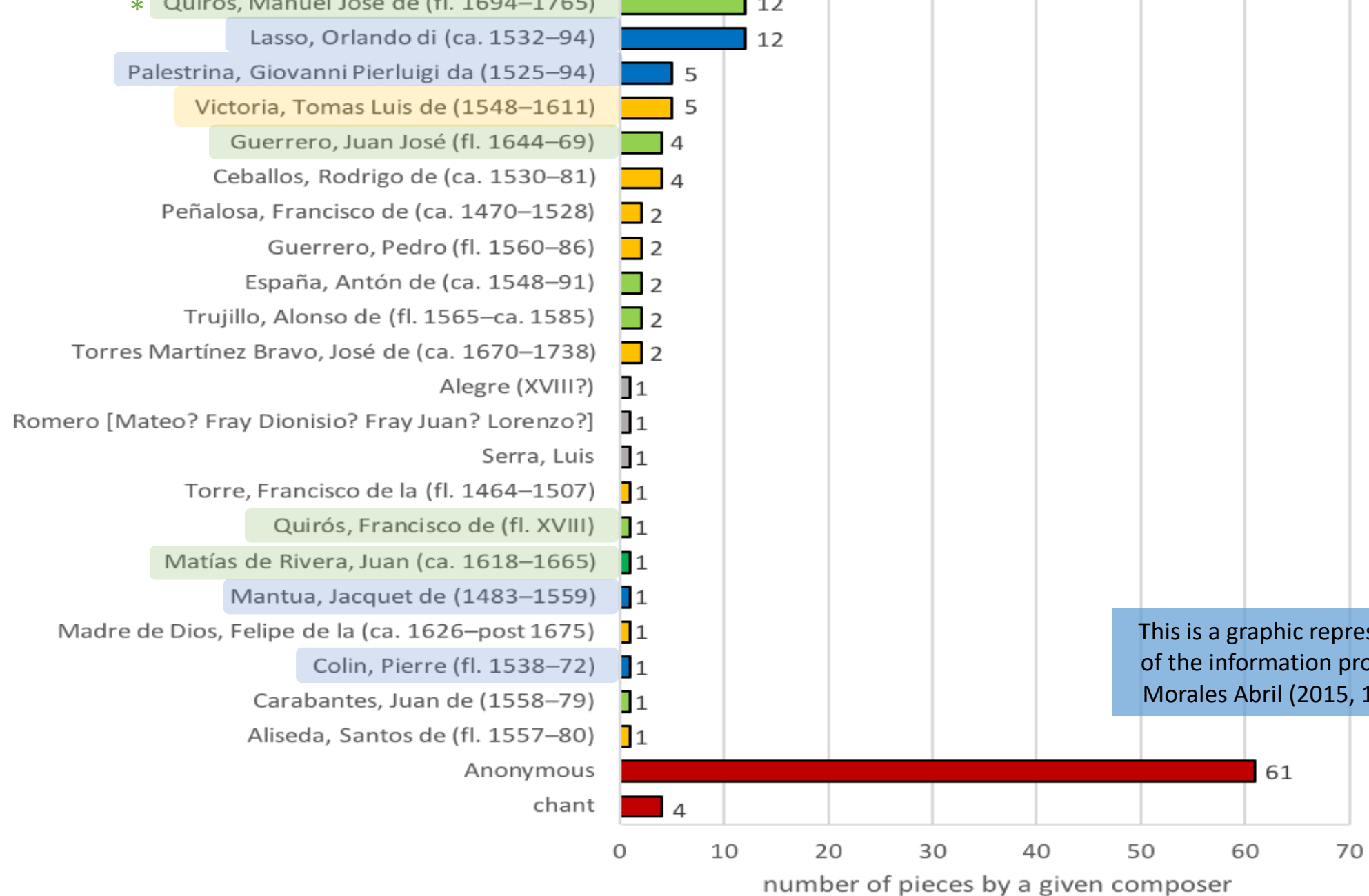
Composers Represented in the GuatC 1–6 Choirbooks



KEY:

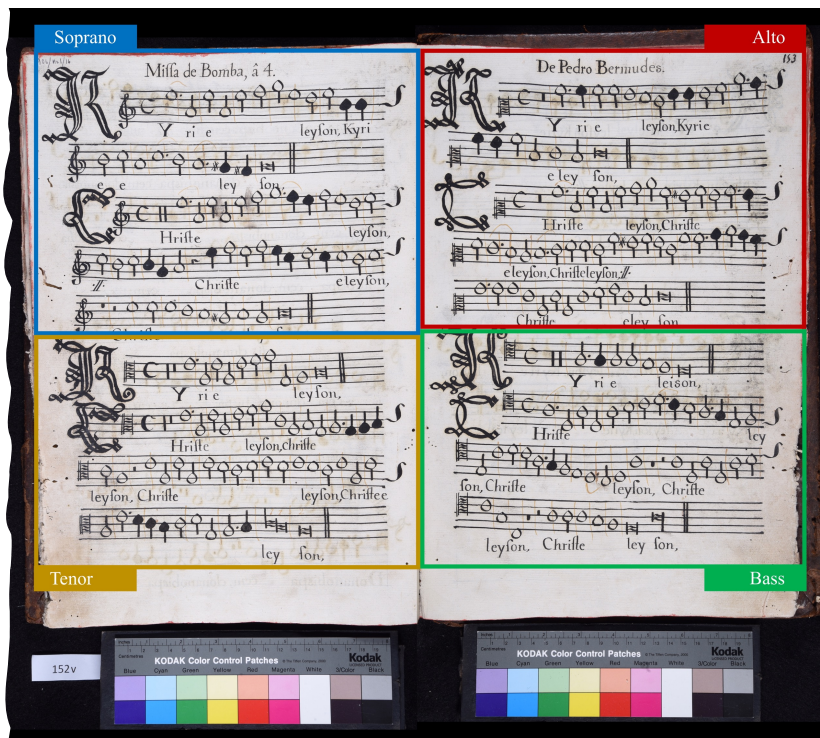
- Iberian Peninsula (never set foot in the Americas)
- Active in the Viceroyalty of New Spain
- Other European composers

This is a graphic representation of the information provided by Morales Abril (2015, 119–120)



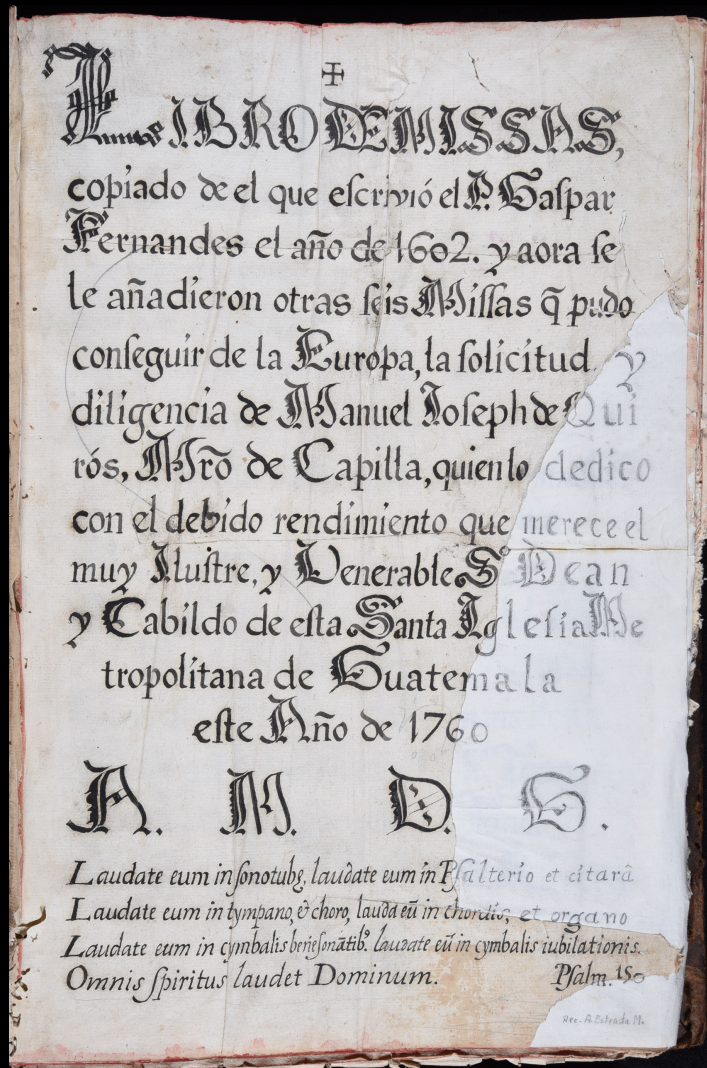
This is a graphic representation of the information provided by Morales Abril (2015, 119–120)

The GuatC Collection



Collection of sacred music

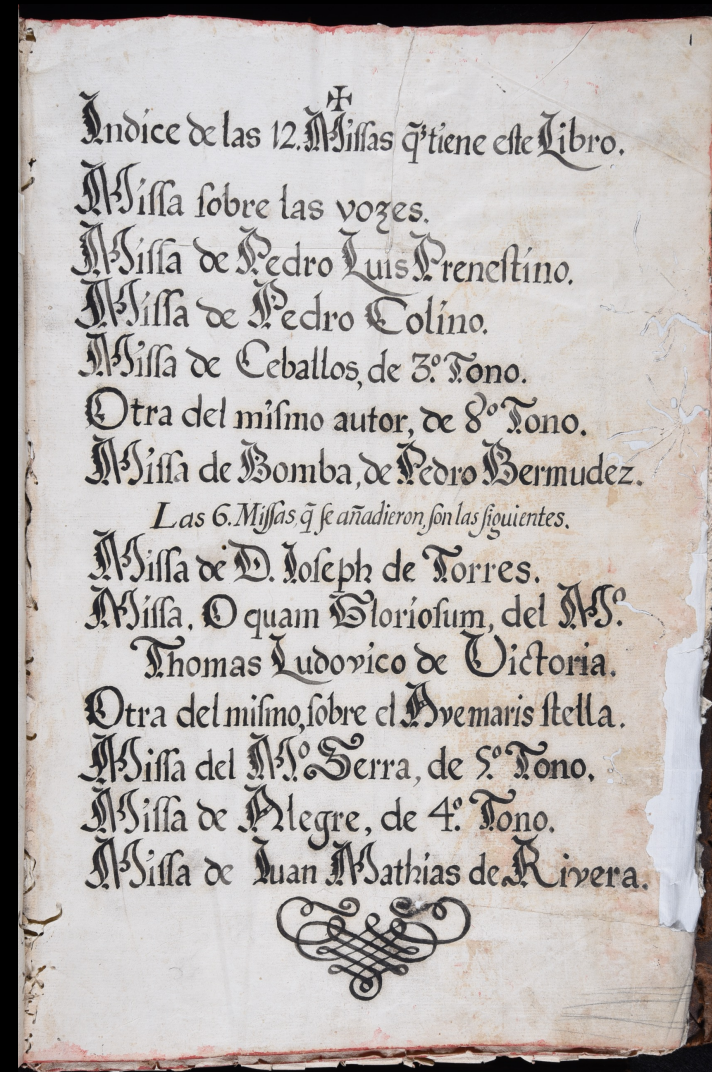
- **GuatC 1:** Book of Masses
- **GuatC 2:** Book of hymns and Magnificats
- **GuatC 3:** Miscellaneous sacred contents
- **GuatC 4:** Book for Holy Week and Salve services
- **GuatC 5:** Miscellaneous sacred contents
- **GuatC 6:** Office of the Dead



GuatC 1, folio 0r

GuatC 1 Book of Masses

- 12 masses
- 15 polyphonic liturgical pieces (eighteenth-century addition)



GuatC 1, folio 1r

Acknowledgements

- **David Rizo, Karen Desmond, and Juliette Regimbal** who made possible for MuRET and the MP-Editor to communicate with one another
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